



Triple Tres (333)

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AI / LLM

MCP Protocol Agentic Loops

LLM Orchestration

MCP Primitives Multi-Agent

Claude API Tool Use API

RAG · pgvector SSE Transport

n8n Flows

BACKEND / INFRA

TypeScript Node.js Python

PostgreSQL SQLite Docker

Railway REST APIs Git Linux

FRONTEND

Next.js React Tailwind

TypeScript shadcn/ui

EDUCATION & CERTS

Model Context Protocol: Advanced Topics
Anthropic · Oct 2025

Claude Code in Action
Anthropic · Sep 2025

Introduction to Model Context Protocol
Anthropic · Sep 2025

AI Fluency: Framework & Foundations
Anthropic · 2025

n8n Self-Hosted for Enterprises
n8n · Nov 2025

Building Scalable Agentic Systems
DataCamp · 2025

B.A. Graphic Design
UAA · 2013

LANGUAGES

Spanish Native

English Professional

AI SOFTWARE ENGINEER · MCP SERVER DESIGN · LLM AGENTIC SYSTEMS

Self-taught engineer shipping **production LLM systems** — from ground-up **MCP server design** (tools, resources & prompts primitives; SSE + HTTP Streamable transport) to a **3-tier multi-agent orchestration** layer with 10 domain-specialized agentic loops. Background in graphic design, crypto analytics, and full-stack dev. Owns the full stack from Node.js backends to React 19 frontends.

EXPERIENCE

Founder & Lead AI Engineer · DISAI

Jun 2025 – Present

- **43-Tool MCP Server:** Architected and deployed a MCP server (TypeScript/Node.js, Railway) covering all three primitives — **Tools** for CFDI 4.0 fiscal ops, **Resources** for live SAT catalog injection (batch-loaded per agent domain), **SSE + HTTP Streamable** dual transport, multi-tenant CSD management. EFOS/EDOS blacklist sync, PAC stamping. **100k+ invoices/min at <2ms latency.** MCP TypeScript Railway
- **3-Tier Agentic Orchestration:** Haiku router (classifies 10 fiscal domains + query intent in ~100ms) → 10 domain **DomainAgents** (Sonnet, Anthropic native tool_use, real agentic loops up to 6 iterations, parallel tool calls, self-correction on failure) → **Expert Registry** with scoped tool allowlists (5–8/domain) + live MCP Resource injection. Zero hallucinations. Claude API Tool Use Multi-Agent
- **Vector DB & MCP Resource Injection:** pgvector (PostgreSQL 16) for semantic SAT catalog search. MCP Resources pre-loaded into each agent's context window before the first LLM call, eliminating hallucinated catalog codes without RAG query overhead. pgvector PostgreSQL
- **Self-Hosted AI Infra:** Deployed self-hosted n8n with direct MCP client integration for SAT tool execution. Replaced Zapier/Make — **80% cost reduction**, 400+ integrations, full data ownership. Keep-alive crons via Railway. n8n Docker
- **Streaming React Frontend:** Next.js 16 + shadcn/ui chat interface with SSE event pipeline — Activity Log, Thinking Steps, Skeleton bubbles for 30–60s async LLM waits. Compliance dashboard, EFOS monitor, CSD status, driver.js onboarding, per-call savings tracking. Next.js React 19 Tailwind

Core Contributor / Lead Developer · RetailDAO

Jun 2024 – Jan 2026

- **Financial Analytics Terminal:** Full-stack "Bloomberg-style" dashboard aggregating real-time data from Federal Reserve, on-chain APIs and crypto exchanges via WebSockets. Rate-limiting, memory caching, and graceful degradation for a self-healing low-latency system. React WebSockets APIs
- **Smart Contract Audits & Documentation:** Led internal audits and built a token distribution visualization platform. Mitigated investor risk and drove a **30% increase in token valuation** to all-time highs.

Crypto Research Analyst · Coinsider

Mar 2024 – Sep 2025

- **Data-Driven Intelligence:** Delivered 25+ peer-reviewed analytics reports (1,200+ pages) — SWOT analysis, investor risk models, and macroeconomic forecasting for Web3 ecosystems. Translated dense technical data into content generating +70k views.

KEY PROJECTS

SAT-MCP Server

43-tool MCP server for Mexican fiscal compliance. Full primitive set: tools, resources, prompts, elicitations, tasks + 8 MCP-UI mini-apps for in-chat fiscal rendering. SSE + HTTP Streamable. CFDI 4.0 lifecycle, EFOS/EDOS sync, PAC stamping, multi-tenant CSD. 3,960 tests · 91%+ coverage.

TypeScript MCP SDK Node.js Railway
SQLite

n8n AI Automation Stack

Self-hosted n8n with direct MCP client integration for SAT tool execution. Specialized sub-agents per CFDI type, 400+ integrations, context bundle architecture.

n8n MCP LLM Agents Docker

DISAI_Conta – 3-Tier Agent System

Haiku router + 10 domain DomainAgents (Sonnet, native tool_use, 6-iter agentic loops) + Expert Registry with scoped allowlists and live MCP Resource injection. Streaming SSE UI for Mexican SMEs.

Next.js 16 Claude API Tool Use MCP Client
shadcn/ui

Homelab AI Infrastructure

PostgreSQL 16 + pgvector for semantic SAT catalog search. Self-hosted Gitea, static IPs, Samba share. Powers the MCP Resource injection pipeline in production.

PostgreSQL pgvector Linux Gitea Docker

